



TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.301

Tire Durability

Test Carrier:

Vehicles in Convoy

Testing Facility:

Street

Status (Porsche):

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Author:

M.Bauer

Translated:

H.Irmler

(see target data sheet / "Zieldatenblatt")

1 Purpose

Testing the wear behavior of tires

2 Field of Application

This test method specifies a procedure for assessing the wear behavior of tires on the vehicle.

Dieses Prüfverfahren ist gültig für:

- Approval Tests

3 Definition of Terms

N/A

4 Responsibility and Scope of Testing

Approval Tests:

The test for approval is performed and documented by the tire manufacturer. The results are made available to Porsche. The Excel evaluation "F.360.301 Tires - Tire durability" template must be used.

Retesting:

Retesting of clearance tires can be arranged by the Porsche Technical Department if required.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

- Tire tread depth in the main tread grooves and in the shoulder area (inside and outside)
- Regular measurement of all tires at mileage levels New, 2,500, 5,000, 7,500, 10,000, 12,500 and 15,000 km
- A tire evaluation must be performed with regard to comfort and noise at 7,500 km and 15,000 km by a test driver known to Porsche from the tire industry
- Description of the track condition (dry, wet)
- Recording of ambient temperatures
- Checking the tire pressure at every shift change
- Recording of longitudinal and lateral accelerations, velocities and determination of frequencies
- If an alternative abrasion route is used, this must be discussed with Porsche in advance of the tests. For this purpose, at least a current correlation to the reference track ("Walsrode") must be available.

5.1.2 Installation of the transducers

Measurement technology for recording acceleration and speed

5.1.3 Data Processing

N/A

5.2 Test Conditions

- Customer tire wear with a mileage of at least 15,000 km
If the wear condition of summer tires has not yet reached 50% of the tread depth after this mileage (25% for all-season tires), the mileage must be increased after consultation with Porsche
- Test and reference track is the special PAG tire wear endurance course "Walsrode"
- The ambient temperature should be $20^{\circ}\text{C} \pm 10^{\circ}\text{C}$ for summer tires and all-season tires
- Ambient temperatures should be $0^{\circ}\text{C} \pm 10^{\circ}\text{C}$ for winter tires

5.3 Test Carrier / Test Set-Up

- Test track near Walsrode:
Proportions:
BAB: approx. 60%
Country Road: approx. 30%
Winding Country Road: approx. 10%
- Determination of the lane speed for all series
BAB:
Average speed: 110 km/h $\pm$ 10 km/h
max. high-speed: 180 km/h
Country Road:
Average speed: 50 km/h $\pm$ 10 km/h
max. high-speed: 100 km/h
- At least two identical vehicles (same technical equipment and chassis setup) are tested simultaneously for a wear endurance run. The definition of the respective vehicle is made by Porsche before the start of the test.
- If only one vehicle is available for the wear measurement, the reference tire and comparison tire must be swapped after each lap or shift. Care must be taken to ensure that the reference and comparison set are subject to the same boundary conditions as far as possible (weather, temperature, traffic thickness, etc.).
- Vehicles and wheel loads of the vehicle as well as tire inflation pressures are to be set according to the target data sheet (partially loaded condition).
- The chassis are measured (weights, wheel position, etc.) and, if necessary, brought up to standard.
- Rim width, diameter and contour at the rim seat of the test rim must correspond to the respective original Porsche wheel (see target data sheet).
- The tires are assigned to the respective wheel position and marked. The wheel positions must not be changed.
- The tread depth measuring points should be defined and marked on the tire as follows:
In each longitudinal profile groove with wear indicator (TWI) with at least three measuring points above the circumference.
At a suitable location on the outer and inner shoulder area with three measuring points above the circumference.

5.4 Test Procedure

- One vehicle is fitted with reference tires, the others with test tires.
- The tread depth measuring points are marked on the tire as defined in 5.3.
- The chassis are measured (weights, wheel positions, etc.) and the weights are brought to the same level if necessary.
- Before drive start, check the tire pressure.
- The vehicles drive directly behind each other on the test track in the same gear, preferably in gear "D".
- The driver, vehicle guidance and tire set (reference set, test set) must be changed at sensible

intervals so that at the end of the test the distances covered with all combinations are as similar as possible.

- At regular intervals (see below), the tread depths at the measuring points and the distance covered with the tires are logged.
- The test is finished as soon as
 - a tire reaches the mileage target value and the tread depth is at least 1.6 mm or
 - the tire has reached 15,000 km mileage and, for summer tires, at least 50% of the tread depth of the new condition (25% for all-season tires). In this case, a regression line is placed through the measured values in the total mileage diagram and linearly extrapolated up to 1.6 mm (see example in Excel worksheet "F.360.301 Tires - Tire durability"). Tread depth measurement values must be available at least at mileages as specified in the parameter table.

6 Documentation

6.1 Data Analysis

- Evaluation of the acceleration, its frequency and the average speed per route section
- The tread depth measurement values of a longitudinal tire groove per measurement are averaged
- For each tire, the parameter "km mileage" is used to create a "Tread depth over longitudinal grooves" diagram (see example in Excel worksheet "F.360.301 Tires - Tire durability")
- The second graph shows the total mileage diagram (tread depth versus km mileage, "specific abrasion FA or specific abrasion RA"). It is based on the measured values of the measuring points with the lowest tread depth at the end of the test.
- For tires that are not worn down, linear extrapolation is performed up to a tread depth of 1.6 mm as follows:
- Expected mileage = (average tread depth NEW - 1.6 mm) / spec. wear (mm/1,000km) * 1000 (see example calculation line in Excel worksheet "F.360.301 Tires - Tire durability").

6.2 Presentation of the results

In the form of a test report (see example Excel worksheet "F.360.301 Tires - Tire durability")

6.3 Requirements

The tire must achieve the values specified in the target data sheet.

7 Applicable Documents

Excel worksheet "F.360.301 Tire - Tire durability")

8 Distributor

EFF